Hybrid Vision–Language Approach for Concept-Based Medical Caption Generation–Concept Detection

Senior 1 Graduation Project

Prepared to obtain the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science

at the **Faculty of Artificial Intelligence Engineering**, Syrian Private University

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SUPERVISOR CERTIFICATION

I certify that the preparation of this project, entitled "Hybrid Vision–Language Approach for Concept-Based Medical Caption Generation–Concept Detection", prepared by "Bashar Alsaleh", was carried out under my supervision at the Faculty of Artificial Intelligence Engineering, Syrian Private University, in partial fulfillment of the requirements for the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science.

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I am equally grateful to the teaching staff of the Faculty of Artificial Intelligence Engineering for providing a stimulating academic environment and for equipping us with the theoretical foundations and practical skills needed to complete this research. Their lectures, discussions, and encouragement contributed significantly to my growth as an artificial intelligence engineering student.

I also extend my appreciation to my family and colleagues for their patience, motivation, and moral support. Finally, I would like to acknowledge the organizers of ImageCLEFmedical Caption 2026 for providing a challenging benchmark and a valuable opportunity to evaluate our work within a competitive scientific framework.

Dedication

I dedicate this work to my family, whose encouragement, patience, and unwavering support have always been a source of strength and motivation. I also dedicate it to my teachers and mentors, who inspired in me a deep interest in scientific inquiry, responsible innovation, and the transformative role of artificial intelligence in healthcare.

This project is also dedicated to every student who strives to connect academic learning with practical impact. I hope that this work, despite its limitations, reflects a sincere effort to contribute to the field of medical image understanding and to the broader goal of building intelligent systems that can support healthcare applications.

Abstract

This thesis addresses the concept detection task of ImageCLEFmedical Caption 2026, where the goal is to predict a set of clinically meaningful UMLS concepts for each radiology image. The task is formulated as a multi-label classification problem and constitutes a crucial building block for downstream medical caption generation and multimodal clinical understanding. However, it remains difficult because of the long-tailed label distribution, severe class imbalance, and the subtle visual patterns that characterize many radiological concepts.

To tackle this problem, the work follows a staged experimental strategy that begins with a DenseNet-121 baseline, continues with threshold calibration and cardinality control, then explores a SwinGraphHead design with rare/common separation, and finally proposes a frequency-aware multi-head Swin Transformer model. In the final architecture, the concept vocabulary is partitioned into four frequency groups, each handled by a dedicated classification head with independently calibrated thresholds. The model is trained using Asymmetric Loss, a WeightedRandomSampler, and an exponential moving average of parameters, and it is evaluated through three-fold multilabel stratified validation.

The final proposed model achieved a validation micro-F1 of 0.5441 ± 0.0004 at the micro-opt operating point. A coverage-oriented operating point achieved a validation micro-F1 of 0.5405 ± 0.0009 while improving macro_nz by 26.1% and increasing concept coverage from approximately 65.7 to 105 concepts. The official challenge submission, generated using the final model, achieved an official test score of 0.5725 and a secondary score of 0.9419. In the corrected final ranking of the ImageCLEFmed Caption 2026 Concept Detection Task, our team, basharderar, ranked fourth among the top-performing teams. These findings indicate that frequency-aware decomposition and per-head calibration offer a practical and effective strategy for long-tailed medical concept detection.

Keywords: medical concept detection, multi-label classification, Swin Transformer, long-tail learning, threshold calibration, ImageCLEFmedical Caption 2026.

الملخص

يهدف هذا المشروع إلى معالجة مهمة **اكتشاف المفاهيم الطبية من الصور الشعاعية** ضمن إطار مسابقة **ImageCLEFmedical Caption 2026**، وهي مهمة تصنيف متعددة اللواصق تهدف إلى التنبؤ بمجموعة من مفاهيم **UMLS** ذات الصلة السريرية لكل صورة. وتكمن أهمية هذه المهمة في أنها تشكّل خطوة أساسية نحو بناء أنظمة قادرة على توليد أوصاف طبية دقيقة للصور الطبية، كما أنها تواجه تحديات حقيقية تتعلق بعدم توازن البيانات، والطبيعة طويلة الذيل لتوزع المفاهيم، وصعوبة تمييز الأنماط البصرية الدقيقة في الصور الشعاعية.

يقترح هذا العمل منهجية هجينة تعتمد على الرؤية واللغة بشكل غير مباشر، وذلك عبر التنبؤ بالمفاهيم الطبية التي تمثّل اللبنات الدلالية الأساسية للتوليد النصي اللاحق. ولتحقيق ذلك، تم تطوير عدة نماذج تجريبية بدأت بخط أساس يعتمد على **DenseNet-121**، ثم مرحلة لمعايرة العتبات والتحكم بعدد المفاهيم المتوقعة، تلاها نموذج **SwinGraphHead**، وصولًا إلى النموذج المقترح النهائي القائم على **Swin Transformer** مع بنية متعددة الرؤوس واعية بالتكرار **(Frequency-Aware Multi-Head)**. تقوم هذه البنية بتقسيم المفاهيم إلى أربع مجموعات وفقًا لتكرارها في بيانات التدريب، وتستخدم رأس تصنيف مستقلًا لكل مجموعة مع معايرة مستقلة للعتبات، إضافةً إلى استخدام **Asymmetric Loss** و **EMA** ومعاينة موزونة للتعامل مع مشكلة عدم التوازن.

أظهرت النتائج أن النموذج المقترح النهائي حقق على مجموعة التحقق قيمة **Micro-F1** مقدارها **0.5441 ± 0.0004** عند نقطة التشغيل المثلى، كما حقق عند نقطة تشغيل ذات تغطية أعلى قيمة **Micro-F1** مقدارها **0.5405 ± 0.0009**، مع تحسن في **macro_nz** بنسبة **26.1%** وزيادة واضحة في عدد المفاهيم المغطاة. كما حقق النموذج، عند رفعه على منصة المسابقة، نتيجة رسمية مقدارها **0.5725** ونتيجة ثانوية **(secondary score)** مقدارها **0.9419**. وفي الترتيب النهائي المصحح لمهمة اكتشاف المفاهيم ضمن مسابقة **ImageCLEFmed Caption 2026**، حصل فريقنا **basharderar** على المركز الرابع بين الفرق الأعلى أداءً. وتؤكد هذه النتائج أن تقسيم المفاهيم بحسب تكرارها، مع استخدام معايرة خاصة بكل مجموعة، يمثل اتجاهًا واعدًا لتحسين اكتشاف المفاهيم الطبية في البيئات ذات التوزع طويل الذيل.

الكلمات المفتاحية : اكتشاف المفاهيم الطبية، التصنيف متعدد اللواصق، **Swin Transformer**، عدم توازن البيانات، التوزع طويل الذيل، **ImageCLEFmedical Caption 2026**.

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Symbol / Term	English Equivalent	Meaning / Description
UMLS	Unified Medical Language System	Ontology used to represent medical concepts by unique CUIs.
CUI	Concept Unique Identifier	Identifier assigned to each medical concept in UMLS.
CNN	Convolutional Neural Network	A class of neural networks commonly used for image analysis.
ViT	Vision Transformer	Transformer-based image encoder using patch embeddings.
Swin	Shifted Window Transformer	Hierarchical transformer backbone with local self-attention windows.
BCE	Binary Cross-Entropy	Standard loss for independent binary labels.
ASL	Asymmetric Loss	Multi-label loss designed to better handle class imbalance.
EMA	Exponential Moving Average	Weight averaging strategy used to stabilize training.
CV	Cross-Validation	Repeated train/validation procedure for robust estimation.
F1-score	Harmonic mean of precision and recall	Primary balanced performance metric.
Micro-F1	Micro-averaged F1 score	Global F1 computed over all predictions and labels.
Macro-F1	Macro-averaged F1 score	Average F1 over labels, treating each label equally.
macro_nz	Non-zero macro-F1	Macro-F1 computed over labels with at least one positive prediction.
Precision	Positive predictive value	Fraction of predicted positives that are correct.
Recall	Sensitivity	Fraction of actual positives that are correctly predicted.
TP	True Positives	Correctly predicted positive labels.
FP	False Positives	Incorrectly predicted positive labels.
FN	False Negatives	Missed positive labels.

Threshold	Decision threshold	Probability cut-off used to convert scores to predicted labels.
Coverage	Concept coverage	Number of concepts predicted at least once on validation/test data.
Long-tail	Long-tailed distribution	Highly imbalanced label frequency distribution with many rare concepts.
ROCOv2	Radiology Objects in Context v2	Radiology benchmark dataset used in the challenge.
ImageCLEF	ImageCLEF medical caption task	Shared task benchmark for medical concept and caption prediction.
DenseNet-121	Dense convolutional network	Baseline CNN backbone used in early experiments.
WeightedRandomSampler	Sampling strategy	Training sampler used to mitigate imbalance.
Top-k	Top-k prediction	Inference strategy selecting the k highest confidence labels.

Chapter 1: Introduction

This chapter introduces the general context of the research and explains the scientific problem addressed in this thesis. It presents the motivation for concept-based medical caption generation, formulates the research problem, defines the main objectives and contributions, identifies the expected beneficiaries, and outlines the organization of the remaining chapters.

1.1 Background and Motivation

Artificial intelligence has become a central tool in medical image analysis, where it supports detection, classification, segmentation, report generation, and decision support. Among these tasks, the automatic recognition of clinically relevant concepts from medical images is particularly important because it can bridge visual understanding and textual interpretation. In radiology, this capability helps convert an image into a structured set of semantics—such as anatomical regions, modalities, findings, and devices—that are closely related to how physicians describe images in reports.

Within the ImageCLEFmedical Caption challenge, concept detection serves as a foundational sub-task for concept-based medical caption generation. Rather than generating a full sentence directly, the system first predicts a set of medical concepts, which can later act as the semantic building blocks of a caption. This formulation is attractive because it provides clinically interpretable intermediate outputs and allows the research to focus on semantic content before full language generation.

The increasing use of deep learning in medical image analysis has created new opportunities for building systems that can extract structured information from visual data. However, medical images differ substantially from natural images: they are often grayscale, modality-dependent, clinically specialized, and difficult to interpret without domain knowledge. As a result, a model that performs well on general image classification cannot be assumed to perform equally well on radiology concept detection without careful adaptation and validation.

Medical caption generation is usually viewed as a vision–language problem because the desired output is textual. Nevertheless, direct generation of a caption from an image can be difficult to evaluate and may hide semantic errors. A concept-based strategy decomposes the problem into an intermediate semantic stage: the system first predicts relevant medical concepts, then these concepts can support later text generation. This makes the intermediate output more interpretable and easier to evaluate objectively.

In this project, the concept detection task is treated as the central stage of a broader Hybrid Vision–Language Approach. The visual component extracts information from radiology images, while the language-oriented component appears through standardized UMLS concepts that represent the semantic meaning of the image. This combination provides a bridge between raw pixels and medically meaningful textual descriptions.

The importance of this task is not limited to challenge participation. A reliable concept detector may support image retrieval, report indexing, weakly supervised report generation, dataset organization, and educational search. In each of these applications, the ability to map an image to clinically interpretable concepts is valuable because it makes the system output more transparent than a black-box caption alone.

1.2 Scientific Problem of the Research

The scientific problem addressed in this thesis is the automatic detection of multiple UMLS-aligned medical concepts from radiology images under a highly imbalanced long-tailed label distribution. The challenge is not simply to optimize a single aggregate score, but to balance strong micro-level performance with broader coverage of rare but clinically important concepts. Standard CNN-based models and global thresholding schemes often favor frequent concepts and underperform on rare labels, which reduces semantic richness and limits downstream captioning utility.

The scientific problem can therefore be formulated as follows: given a radiology image, predict the set of UMLS concepts that best describe its clinically relevant visual content. This problem is challenging because the output space is large, the number of positive labels per image is small, and the distribution of concepts is extremely imbalanced.

A central challenge is the long-tail distribution of labels. In such a distribution, a small number of concepts dominate the dataset, while many concepts appear rarely. A model trained naively on this distribution may learn to predict common labels very well while almost ignoring rare labels. This behavior may still lead to a reasonable micro-F1 score because common labels dominate the metric, but it limits the semantic richness of the predictions.

Another challenge is the heterogeneity of the visual data. The dataset includes different radiological modalities and anatomical regions. A single model must therefore learn visual patterns across a broad range of image appearances. This motivates the use of a strong backbone such as Swin Transformer, which can model hierarchical visual features and contextual relationships.

The problem also includes an inference-level component. Even when a model outputs useful probabilities, the final set of predicted concepts depends on thresholding. Therefore, the thesis treats threshold calibration as part of the methodology rather than as a minor post-processing detail.

1.3 Research Objectives

- To study the concept detection task of ImageCLEFmedical Caption 2026 as a clinically relevant multi-label classification problem.
- To establish a strong CNN baseline and analyze the effects of threshold selection and prediction cardinality.
- To investigate transformer-based alternatives, especially a Swin-based frequency-aware architecture.
- To design a final model that improves long-tail behavior while preserving competitive micro-F1 performance.
- To compare the proposed model with intermediate experimental variants and with representative previous studies.

The objectives are organized around both modeling and analysis. The first goal is to understand the dataset and quantify its main difficulties. The second goal is to construct a baseline that is strong enough to be meaningful. The third goal is to improve the model design and calibration strategy in a way that directly addresses long-tail behavior.

The project also aims to produce a scientifically traceable sequence of experiments. This is important for a graduation thesis because the final result alone does not explain the research process. By documenting each experimental stage, the thesis demonstrates how the final approach was reached and why specific design choices were adopted.

1.4 Research Contributions

- A structured experimental progression from DenseNet baselines to a final frequency-aware Swin multi-head model.
- An explicit analysis of threshold calibration and concept cardinality in long-tailed medical concept detection.
- A final proposed model that partitions concepts into four frequency bins and calibrates thresholds per head.
- An evaluation framework that contrasts micro-opt and coverage-opt operating points, providing a richer interpretation of model behavior.
- A successful official submission to the ImageCLEFmed Caption 2026 Concept Detection Task, achieving an official test score of 0.5725 and ranking fourth in the corrected final ranking.

The main contribution is the final Swin Frequency-Aware Multi-Head architecture. The model uses a shared Swin Transformer backbone to extract visual features, but it does not treat all concepts uniformly at the classification stage. Instead, it divides the concept vocabulary into four frequency groups and assigns a dedicated head to each group.

A second contribution is the systematic study of operating points. The thesis distinguishes between a micro-F1 optimized operating point and a coverage-oriented operating point. This distinction is important because a concept detector for caption generation may need to recover a broader set of concepts, not only maximize a single aggregate score.

A third contribution is the complete experimental pipeline, including data loading, preprocessing, label binarization, stratified folds, model training, validation inference, threshold calibration, and official submission generation. This makes the work reproducible and suitable for academic evaluation.

1.5 Beneficiaries of the Research

The beneficiaries of this research include academic researchers working on medical image understanding, students studying applied artificial intelligence in healthcare, and future developers of multimodal medical captioning systems. More broadly, the methodological insights of this thesis may benefit systems that require interpretable semantic tags as intermediate representations.

The direct beneficiaries of this work include researchers and students working on medical image understanding and multimodal learning. The thesis provides a detailed example of how a challenge-oriented AI project can be developed from data exploration to official test submission.

Indirect beneficiaries include future systems that use medical concepts as intermediate representations. Although the current work is not a clinical product, the methodology can support future research in explainable medical image analysis, medical report generation, and concept-based retrieval.

The scope of this thesis is limited to the concept detection part of the broader captioning pipeline. This limitation is intentional. Concept detection is a measurable and interpretable stage, and improving it provides a solid foundation for later caption generation. By focusing on this stage, the project can evaluate the model using objective metrics and analyze the behavior of different label groups.

The work also follows an engineering research methodology. The solution is not evaluated only theoretically; it is implemented, trained, validated, calibrated, and submitted to an external challenge platform. This makes the project closer to a real applied AI research workflow than a purely conceptual design.

The problem is especially suitable for a graduation project in artificial intelligence engineering because it combines several important topics: deep learning, medical image processing, multi-label classification, dataset engineering, long-tail learning, model evaluation, and scientific reporting.

1.6 Organization of the Thesis

The remainder of the thesis is organized as follows. Chapter 2 presents the theoretical background, including multi-label learning, transformer backbones, long-tail learning, and evaluation metrics. Chapter 3 reviews relevant literature and identifies the gap motivating the proposed method. Chapter 4 explains the datasets, the experimental pipeline, the proposed final model, and the training/calibration procedure. Chapter 5 reports the quantitative results and comparisons with previous work. Chapter 6 concludes the thesis and discusses current limitations and future directions.

1.7 Chapter Summary

This chapter presented the research context, the scientific problem, and the main objectives of the thesis. It clarified why medical concept detection is a challenging long-tailed multi-label problem and explained how the proposed work is positioned as an interpretable stage within a broader vision-language captioning pipeline.

Chapter 2: Theoretical Background

This chapter reviews the theoretical foundations needed to understand the proposed approach. It explains medical concept detection as a multi-label learning problem, summarizes relevant visual backbones, introduces Swin Transformer, discusses long-tailed learning and imbalance handling, and defines the evaluation metrics used throughout the thesis.

2.1 Medical Concept Detection as a Multi-Label Classification Problem

Medical concept detection can be formulated as a multi-label classification problem because each radiology image may be associated with multiple semantic concepts simultaneously. These concepts may refer to anatomy, modality, acquisition settings, findings, pathologies, image characteristics, or medical devices. Unlike multi-class classification, where a sample belongs to exactly one class, multi-label learning requires the model to assign a binary relevance decision to each label in the concept vocabulary.

In ImageCLEFmedical Caption 2026, the target labels correspond to UMLS concepts, which makes the task clinically grounded and semantically structured. Such a formulation is valuable because concept-level outputs are interpretable and can be reused by retrieval systems, report-generation pipelines, or downstream language models.

A multi-label classifier differs fundamentally from a multi-class classifier. In a multi-class setting, a softmax layer is typically used because the classes are mutually exclusive. In concept detection, this assumption is invalid: one image can correspond to several concepts at the same time. Therefore, independent sigmoid outputs are used, allowing each concept to be predicted separately.

The target vector for each image is sparse. If the vocabulary contains 2,646 concepts and an image contains only three or four positive concepts, most entries of the target vector are negative. This creates a strong imbalance between positive and negative labels within each training example. A standard loss function may therefore be dominated by easy negative labels unless it is adjusted.

Although the output labels are represented independently, medical concepts are not semantically independent. Certain anatomical concepts, modalities, and findings may co-occur frequently. The present thesis does not explicitly model the full UMLS graph, but the shared visual backbone and the frequency-aware grouping provide partial mechanisms for learning useful shared representations.

2.2 Deep Learning Backbones for Medical Image Analysis

CNNs such as DenseNet-121 remain strong baselines in radiology due to their ability to learn local texture patterns and spatial motifs efficiently. Dense connectivity improves gradient flow and feature reuse, which is beneficial in medical imaging settings where subtle visual cues often matter. However, CNNs may still be limited in capturing long-range contextual relationships unless deeper or specially designed architectures are used.

Convolutional Neural Networks are historically the dominant family of architectures in medical imaging. Their local receptive fields make them effective for detecting texture, edges, and localized patterns. DenseNet-121 is especially popular because dense connections encourage feature reuse and stable gradient flow. This is one reason it was chosen as the first baseline in the project.

However, CNNs have a locality bias. While this bias is useful, it can limit the ability to model long-range relationships unless the network is very deep or uses additional mechanisms. Radiology images may contain spatial relationships that are not confined to a small region, particularly when anatomical context matters. This provides one motivation for exploring transformer-based backbones.

DenseNet remains important in the thesis not because it is the final model, but because it establishes a strong reference point. A final method should not only perform well in isolation; it should improve upon or provide better behavior than a credible baseline.

2.3 Vision Transformers and Swin Transformer

Vision Transformers model images as sequences of patches and rely on self-attention to capture global context. The Swin Transformer improves practicality by restricting self-attention to shifted local windows while building a hierarchical representation. This design offers a favorable compromise between local detail modeling and broader contextual reasoning, which is particularly attractive for radiology images where both subtle local patterns and larger structural relationships are important.

Transformer-based vision models represent images as sequences of patches and use self-attention to model relationships among those patches. This allows the model to capture broader dependencies than a purely convolutional local filter. In medical images, such global context can help distinguish subtle findings and understand anatomical structure.

The Swin Transformer improves the efficiency of standard vision transformers by computing attention within local windows and shifting these windows across layers. This maintains computational feasibility while enabling cross-window information flow. The architecture also builds hierarchical representations, which are useful for visual tasks that require both local and global information.

In the final model, Swin is used as a feature extractor rather than as the entire solution. The frequency-aware heads are added on top of the backbone because the long-tail problem is not solved by changing the backbone alone. This separation between representation learning and label-frequency handling is a key design principle of the proposed approach.

2.4 Long-Tailed Learning and Imbalance Handling

A major difficulty in medical concept detection is the long-tailed nature of the label distribution. A small subset of concepts appears very frequently, whereas many concepts occur rarely. This imbalance makes global optimization difficult because models tend to learn frequent labels more easily and may produce conservative predictions for rare labels. Strategies for handling this issue include sampling schemes, class-sensitive loss functions, threshold calibration, and architectural decomposition of the label space.

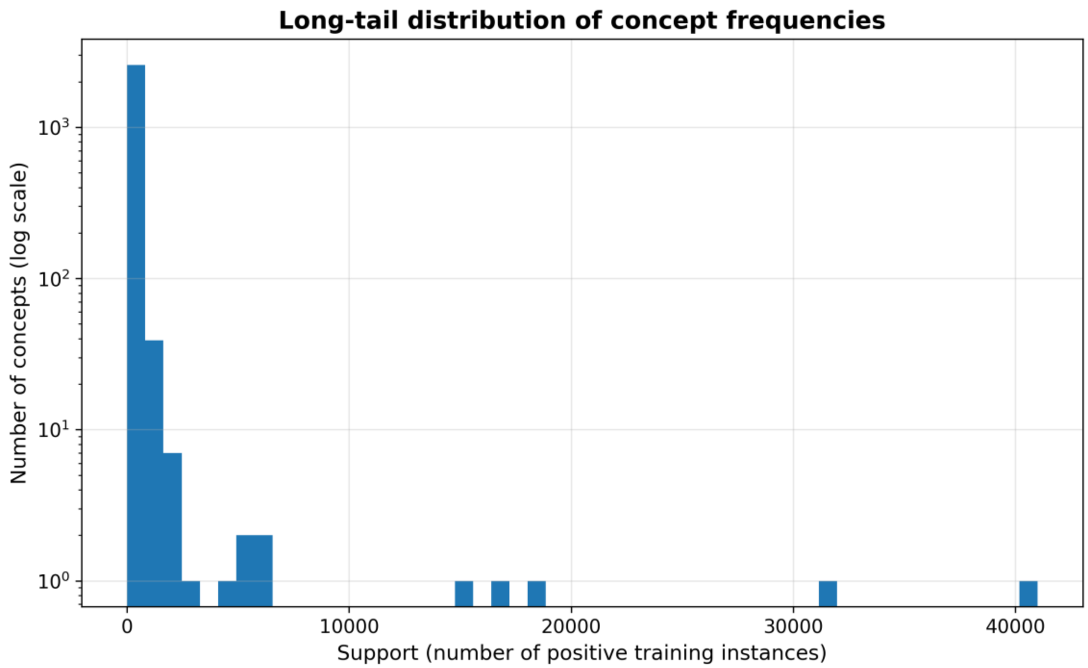

Figure 2.1: Long-tail distribution of concept frequencies.

Long-tailed learning is a common problem in real-world medical data. Clinical concepts are not uniformly distributed because common anatomical and modality labels appear frequently, while rare conditions or specific findings appear much less often. A naïve classifier trained on such data may minimize its loss by focusing on frequent labels.

Several techniques can mitigate this issue. Sampling strategies can increase the presence of rare-label samples during training. Loss functions can reduce the dominance of easy negatives or frequent classes. Architectural strategies can decompose the label space into more manageable groups. Calibration strategies can adjust the decision threshold to better match the behavior of different concept groups.

The proposed method combines these ideas. WeightedRandomSampler addresses data exposure, Asymmetric Loss addresses optimization imbalance, frequency-aware heads address architectural specialization, and per-head thresholds address inference-time decision making.

2.5 Evaluation Metrics and Decision Thresholding

Because the task is multi-label, evaluation must consider the balance between precision and recall. In addition, the choice of threshold used to convert predicted probabilities into binary labels strongly affects the final performance. This is why threshold calibration becomes a central component of the overall pipeline.

$$\textbf{Precision} = \textbf{TP} / (\textbf{TP} + \textbf{FP})$$

Equation (2.1): Precision.

$$\textbf{Recall} = \textbf{TP} / (\textbf{TP} + \textbf{FN})$$

Equation (2.2): Recall.

$$\textbf{F1} = 2 \times \textbf{Precision} \times \textbf{Recall} / (\textbf{Precision} + \textbf{Recall})$$

Equation (2.3): F1-score.

$$\hat{y}_i = \sigma(z_i)$$

Equation (2.4): Sigmoid prediction for multi-label classification.

$$L_{BCE} = - \Sigma_i [\ y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)\ ]$$

Equation (2.5): Binary cross-entropy loss.

$$L_{ASL} = - \Sigma_i [\ y_i (1-\hat{y}_i)^{\gamma+} \log(\hat{y}_i) + (1-y_i)\ p_i^{\gamma-} \log(1-p_i)\ ]$$

Equation (2.6): Asymmetric loss for multi-label classification.

$$\textbf{Micro-F1} = 2\ \textbf{TP} / (2\ \textbf{TP} + \textbf{FP} + \textbf{FN})$$

Equation (2.7): Micro-F1 score.

Evaluation in multi-label classification must consider several complementary metrics. Precision indicates how reliable the predicted concepts are, while recall indicates how complete the predicted concept set is. F1-score balances the two. In a sparse setting, increasing recall often decreases precision, so the chosen operating point must match the intended use.

Micro-F1 is useful as a global metric because it aggregates all decisions before computing the score. However, it can be dominated by frequent labels. Macro-F1 gives each label equal importance, which makes it more sensitive to rare labels. The thesis also reports macro_nz, a practical variant focused on labels with non-zero behavior, and coverage, which counts how many concepts are predicted at least once.

Threshold calibration is therefore a central part of evaluation. A model that looks weak under a default threshold may become stronger after calibration, and a model that maximizes micro-F1 may not provide the richest semantic coverage. This is why both micro-opt and coverage-opt results are reported.

2.6 Additional Theoretical Considerations

A practical medical concept detector must handle not only visual feature extraction but also the semantic structure of the label space. Unlike ordinary object categories, UMLS concepts belong to a controlled medical vocabulary. This means that the output labels can be connected to higher-level medical knowledge, even if the current model does not explicitly use those connections during training.

Another theoretical issue is the difference between image-level evidence and concept-level annotation. Some concepts may be directly visible in the image, while others may be contextual or indirectly associated with image type, body region, or acquisition conditions. A model may therefore learn both visual cues and dataset-level correlations. This is common in large medical datasets and should be considered when interpreting results.

The sparse multi-label setting also creates a difficult optimization landscape. Most labels are negative for most images. If all negative labels are treated equally, the model may spend most of its capacity learning to reject labels rather than learning subtle positive evidence. This is one reason Asymmetric Loss is appropriate: it changes the relative contribution of positive and negative examples.

From a modeling perspective, the final approach can be interpreted as a form of label-space decomposition. Instead of training one very large classifier head for all concepts, the label set is divided into groups based on frequency. This does not reduce the vocabulary size; rather, it organizes the prediction problem in a way that better reflects the observed data distribution.

Theoretically, threshold calibration can be viewed as selecting a point on the precision–recall curve. Different thresholds represent different clinical or application preferences. A strict threshold emphasizes precision and may be suitable when false positives are costly. A more permissive threshold increases recall and may be suitable when downstream systems can filter or use additional context.

This thesis therefore treats the final system as a combination of representation learning, imbalance-aware optimization, and calibrated decision making. All three components are required: a strong backbone without calibration may underperform, a calibrated weak model may lack visual discrimination, and a long-tail-aware architecture without a suitable loss may still be dominated by frequent labels.

2.7 Chapter Summary

This chapter reviewed the theoretical concepts underlying the thesis, namely multi-label learning, modern deep visual backbones, long-tail imbalance, and evaluation metrics. These foundations motivate the transition from a straightforward DenseNet baseline to a more structured frequency-aware transformer solution.

Chapter 3: Related Work

This chapter reviews the most relevant studies and techniques related to medical image classification, medical concept detection, long-tailed learning, and vision-language modeling. It also identifies the research gap that motivates the proposed frequency-aware approach.

3.1 Previous Studies in Similar Domains

The literature relevant to this thesis spans foundational CNN-based radiology classifiers, ontology-aligned medical labeling resources, challenge-oriented concept detection studies, and recent medical vision-language methods. Together, these studies define the scientific context in which our problem is situated.

CheXNet introduced DenseNet-121 as a powerful CNN baseline for chest X-ray analysis and demonstrated that deep convolutional models can achieve high radiological classification performance. CheXpert subsequently expanded this perspective by explicitly addressing uncertainty in report-derived labels. Although these studies do not solve the ImageCLEF concept detection task directly, they provide strong conceptual and methodological baselines for multi-label radiology prediction.

PadChest is highly relevant as a dataset and labeling reference because it emphasizes ontology-aligned annotations and clinically meaningful label structures. In our context, this is important because ImageCLEF concept detection uses UMLS concepts rather than arbitrary labels, making ontology-aware thinking essential even when the learning backbone is purely visual.

Among studies most closely aligned with our task, Santamato et al. (2025) and Sahni et al. (2025) are especially important. Santamato et al. studied ImageCLEF-aligned radiology concept prediction and highlighted the substantial performance gap between reduced label subsets and the complete long-tail concept space. Sahni et al. evaluated several pretrained CNN backbones on ROCOv2-based challenge data and offered a practical challenge-style experimental template, reporting F1 results around the mid-0.4 range under their setup.

Additional support comes from representation-learning research such as ConceptCLIP and MedBLIP. These studies do not directly solve our task under the same protocol, but they motivate concept-aware representation learning and show that semantically aligned medical vision-language models can improve downstream understanding.

Table 3.1: Comparative summary of representative related studies.

Study	Dataset / Setting	Model	Reported result	Strengths	Limitations
CheXNet (2017)	ChestX-ray14	DenseNet-121	Pneumonia F1=0.435	Strong CNN baseline and clinician comparison	Single-modality, not ontology-centered
CheXpert (2019)	CheXpert	CNN baselines	AUROC up to 0.941	Robust uncertainty handling benchmark	AUROC-focused and chest X-ray specific
PadChest (2020)	PadChest	Ontology-aligned label pipeline	Label validation Micro-F1=0.93	Clinically meaningful ontology mapping	Mainly a dataset/labeling resource
Santamato et al. (2025)	ImageCLEF concept prediction	DenseNet121 / ResNet101	F1≈0.26 on 3047 concepts	Challenge-aligned concept detection study	Performance drops sharply on full long-tail space
Sahni et al. (2025)	ROCOv2 / ImageCLEF	ResNet50 / DenseNet121 / Inception variants	F1 up to 0.4686	Practical challenge-style baselines	Reduced label spaces and limited long-tail robustness
ConceptCLIP (2025)	Medical VLM benchmarks	Contrastive VLM	Avg AUC≈70.82	Concept-aware representations	Not directly comparable to ImageCLEF F1
MedBLIP (2025)	ROCO-based captioning	BLIP-based VLM	Improved captioning metrics	Supports multimodal semantic alignment	Indirect to Task 1 concept detection

3.1.1 CheXNet and CNN-Based Multi-Label Radiology Prediction

CheXNet is one of the foundational deep learning works in chest X-ray classification. It used DenseNet-121 to predict thoracic diseases from the NIH ChestX-ray14 dataset. Its relevance to this thesis lies in demonstrating that DenseNet-style CNNs can serve as strong multi-label baselines in radiology.

The reported pneumonia F1-score of approximately 0.435 on a radiologist-labeled test set made CheXNet a widely cited reference. Nevertheless, the task setting is much simpler than ImageCLEFmedical Caption concept detection because CheXNet predicts fourteen labels, whereas our task contains thousands of UMLS concepts.

For this thesis, CheXNet supports the choice of DenseNet-121 as Experiment 1. It also demonstrates the importance of comparing deep learning predictions with meaningful clinical labels. Its main limitation for our work is its chest X-ray focus and its relatively small label space.

3.1.2 CheXpert and Label Uncertainty

CheXpert expanded large-scale chest radiograph learning by explicitly addressing uncertain labels extracted from radiology reports. This is relevant because medical image labels generated from reports or concept extraction pipelines may contain uncertainty and noise.

CheXpert evaluated multiple uncertainty-handling policies and reported high AUROC values for several observations. However, the benchmark is AUROC-centered, while ImageCLEFmedical Caption concept detection emphasizes F1-style evaluation and final predicted concept lists. The study therefore informs our understanding of medical label uncertainty but is not directly equivalent to our task.

3.1.3 PadChest and Ontology-Aligned Labels

PadChest is important because it emphasizes structured, clinically meaningful labels mapped to medical ontologies. This aligns with the UMLS-based nature of ImageCLEF concepts. In concept detection, the label is not merely a class name; it is a standardized identifier that can connect to broader medical knowledge.

PadChest reported high validation quality for its label extraction pipeline, but it should be interpreted mainly as a dataset and labeling contribution rather than an image classifier benchmark. Its relevance to this project is that it supports the importance of ontology-aligned concept spaces.

3.1.4 Vision Transformers and Swin Transformer

Vision Transformers introduced self-attention to image recognition by treating images as patch sequences. This idea is important for medical imaging because attention can model broader spatial relationships. However, standard ViT can be computationally expensive and often requires large-scale pretraining.

Swin Transformer addresses these limitations through shifted local windows and hierarchical feature maps. This makes it more practical for dense visual recognition and more suitable for medical imaging contexts. The final proposed model uses Swin Transformer because it can capture both local patterns and broader context while remaining computationally feasible.

3.1.5 ImageCLEF 2025 and 2026 Related Concept Detection Studies

Santamato et al. (2025) studied CNN transfer learning for radiology concept prediction in an ImageCLEF-related setting. Their results show a major difference between full-vocabulary concept detection and reduced frequent-label experiments. DenseNet121 transfer learning reached approximately F1=0.26 over thousands of concepts, while the top-20 frequent concept setting achieved much higher scores.

This contrast directly supports our focus on long-tail learning. A model that performs well on the most frequent labels may still be weak on the realistic full-vocabulary task. Therefore, the final proposed method uses frequency-aware modeling rather than limiting the vocabulary to common concepts.

Sahni et al. (2025) evaluated several pretrained CNN backbones on ROCOv2-based ImageCLEFmedical Caption data, including ResNet50, DenseNet121, InceptionV3, and Inception-ResNetV2. Their reported F1 values around the mid-0.4 range provide a practical challenge-style comparison point for our work.

3.1.6 ConceptCLIP, MedBLIP, and Vision–Language Learning

ConceptCLIP and MedBLIP represent a newer direction in medical AI: aligning images with medical text or concepts during pretraining. These approaches are relevant to the broader title of the project because they connect visual representation learning with semantic language representations.

The present thesis focuses on the concept detection stage rather than full caption generation. However, concept detection is a natural bridge toward captioning because the predicted CUIs can later be used as structured semantic inputs for a language-generation model. This relationship justifies the hybrid vision–language framing while keeping the current scope scientifically focused.

3.1.7 Comparative Interpretation of the Literature

The reviewed literature shows a gradual evolution from CNN-based disease classification toward more semantically structured and multimodal medical image understanding. CheXNet and CheXpert demonstrate the strength of supervised CNN classifiers, but their output spaces are relatively limited and mostly chest X-ray focused.

PadChest shifts the focus from pure classifier performance to clinically meaningful label organization. This is highly relevant because concept detection quality depends not only on visual classification but also on the consistency and usefulness of the target concepts.

Transformer-based approaches introduce a different representation paradigm. Instead of relying only on local convolutional filters, attention-based backbones can capture longer-range visual dependencies. This is particularly valuable when findings are spatially distributed or when anatomical context is required.

Recent ImageCLEF-related works demonstrate that the main challenge is not simply building a classifier, but scaling concept detection to thousands of labels. The gap between top-frequency results and full-vocabulary results shows that long-tail behavior is a central unresolved issue.

Vision–language models such as ConceptCLIP and MedBLIP suggest a future direction, but they do not remove the need for accurate concept detection. In fact, a concept detector can complement VLMs by providing explicit, standardized semantic supervision.

The proposed method is positioned at the intersection of these directions: it uses a modern transformer backbone, keeps the concept output space explicit, and introduces a frequency-aware structure to address long-tail imbalance.

3.2 Modern Techniques and Frameworks in the Field

Three families of techniques are especially relevant to the present work. First, transformer-based backbones such as ViT and Swin Transformer provide more powerful context modeling than conventional CNNs. Second, long-tail learning techniques—such as Asymmetric Loss, class-sensitive sampling, and tailored decision thresholds—address imbalance at the optimization and inference levels. Third, multimodal methods such as VLMs and concept-aware contrastive models motivate the broader hybrid vision-language framing of our project, even though the current thesis focuses specifically on the concept detection stage.

Framework-wise, the project relies on PyTorch, timm, scikit-learn-compatible evaluation, and mixed-precision training. These choices are consistent with current deep learning practice and are suitable for reproducible experimentation on a large multi-label radiology dataset.

3.3 Comparative Analysis of Existing Systems

From a system perspective, existing approaches differ in three important dimensions: the label space size they attempt to cover, the degree of explicit imbalance handling, and the extent to which they exploit concept-level structure. Many published baselines perform well on frequent or reduced label sets but show limited robustness on the full long-tail distribution. This observation directly motivates a frequency-aware strategy.

Table 3.2: Comparative analysis of existing systems and studies.

System	Main characteristics	Core technology	Strengths	Weaknesses
CheXNet-style CNN	Strong convolutional baseline; simple end-to-end pipeline	DenseNet / CNN	Stable baseline, efficient training	Limited global context modeling
Challenge CNN baselines	Challenge-oriented reporting and reproducibility	ResNet / DenseNet / Inception	Practical and directly comparable	Still weak on rare labels
Concept-aware VLMs	Semantic alignment and broader representations	Contrastive VLM / BLIP	Potentially stronger semantic features	Indirect and often costly for this task
Our proposed method	Frequency-aware concept decomposition with per-head calibration	Swin + multi-head design	Balances micro-F1 and long-tail coverage	More complex training and calibration pipeline

Modern medical AI increasingly combines vision encoders with semantic supervision. CNNs remain efficient and reliable, while transformer-based models provide stronger context modeling. Loss functions such as Asymmetric Loss and sampling techniques are frequently used when labels are sparse and imbalanced.

In this thesis, these techniques are not used independently. They are integrated into a unified pipeline: Swin provides the visual representation, frequency-aware heads provide label-space decomposition, Asymmetric Loss and sampling address imbalance during training, and threshold calibration controls the final predicted concept set.

The framework used for implementation reflects current deep learning practice. PyTorch provides flexible model development, timm provides pretrained Swin backbones and image transformations, iterative stratification supports multilabel fold creation, and pandas/numpy support dataset analysis and label processing.

Despite the progress shown in previous studies, several limitations remain. First, many classical radiology classification studies use relatively small label sets. Their conclusions cannot be transferred directly to a setting with thousands of concepts. Second, studies that use reduced concept sets can achieve high scores, but such results may not reflect performance on realistic full-vocabulary detection.

Third, some vision–language works focus on representation quality or captioning metrics rather than explicit concept-level F1. While these works are important for future integration, the current challenge requires precise concept prediction. Therefore, a model must be evaluated directly on concept detection behavior.

Fourth, many systems use a single global output head. This design is simple but does not account for the different learning dynamics of rare and common concepts. A rare concept with few positive examples and a common concept with thousands of examples should not necessarily be treated identically at the output and calibration stages.

These gaps motivate the proposed model. The final architecture preserves the full vocabulary, uses a modern transformer backbone, decomposes the label space by frequency, and calibrates thresholds in a way that explicitly considers the trade-off between micro-F1 and coverage.

3.4 Research Gap and Proposed Innovation

The literature review reveals a clear gap. Existing CNN baselines and challenge reports confirm that concept detection is feasible, but they also show that performance degrades when the label space becomes large and imbalanced. Most approaches either emphasize frequent labels or use global decision thresholds that do not account for the very different statistical behavior of rare and common concepts. Moreover, while concept-aware and multimodal methods suggest useful directions, they are not yet directly tailored to this exact challenge setting.

Our proposed innovation is therefore to model the long-tailed concept space explicitly. Instead of using a single output head and a single thresholding rule, the final model partitions concepts into four frequency groups and uses a dedicated classification head for each group. This decomposition is coupled with per-head threshold calibration, enabling the system to preserve strong micro-F1 while broadening semantic coverage. In this sense, the contribution is not merely a backbone replacement, but a structured approach to balancing accuracy and coverage in a clinically meaningful label space.

3.5 Chapter Summary

This chapter reviewed representative studies related to medical image classification, ontology-aligned labeling, ImageCLEF concept detection, and medical vision-language learning. The review showed that existing methods often perform well on frequent or reduced label spaces but remain challenged by large long-tailed concept vocabularies. These observations motivate the frequency-aware decomposition strategy proposed in this thesis.

Chapter 4: Research Methodology

This chapter describes the methodology followed to build and evaluate the proposed system. It explains the data preparation process, the sequence of experiments, the final model architecture, and the training and calibration procedures.

4.1 Chapter Introduction

This chapter presents the research methodology followed in the thesis. It starts with the problem formulation and dataset description, then explains the successive experimental stages, and finally details the architecture, training process, calibration strategy, and analysis tools used for the final proposed model.

4.2 Problem Formulation and Overall Pipeline

Given an input radiology image x, the task is to predict a binary vector y over the medical concept vocabulary. The pipeline begins with image preprocessing and label binarization, followed by model training, validation-time threshold calibration, and operating-point selection. The final output is a set of predicted CUIs for each image.

The pipeline begins with raw dataset inspection. Each image ID is matched to its concept list, and the set of unique CUIs is collected to create a fixed vocabulary. This vocabulary defines the order of the output neurons in every model. Maintaining a consistent label ordering is essential because an output index has meaning only relative to the saved vocabulary.

After vocabulary construction, each image label list is converted into a multi-hot vector. If a concept is present in the image, the corresponding position in the vector is set to one; otherwise it is zero. This transformation allows the use of standard multi-label losses.

The visual pipeline applies resizing and normalization consistent with the selected pretrained backbone. For Swin Transformer, the timm data configuration is used to ensure that image resolution, interpolation, and normalization match the expected training distribution of the pretrained model.

4.3 Datasets and Preprocessing

The development set used in this work contains 116,604 image-label pairs stored as radiology images together with concepts.csv and attribution.csv files. The concept vocabulary contains approximately 2,646 unique CUIs, with an average of about 3.22 labels per image. Labels are transformed using a multi-hot representation, and images are resized to 224×224 according to the timm data configuration of the selected backbone.

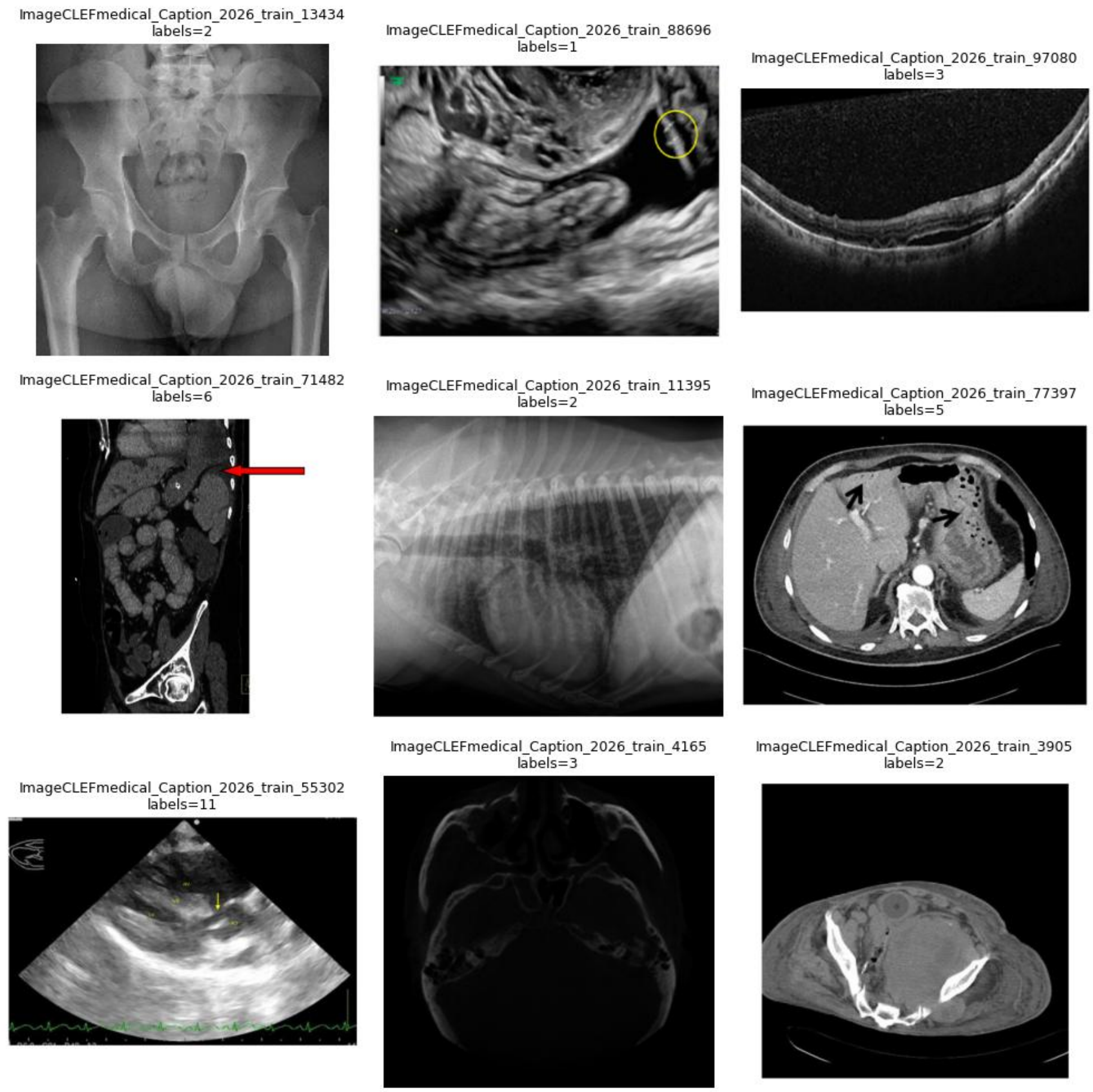

Figure 4.1: Sample radiology images from the development dataset.

Table 4.1: Dataset statistics of the development split used in this work.

Statistic	Value
Number of training/validation images	116,604
Concept vocabulary size	2,646 CUIs
Average labels per image	3.22
Image input size	224 × 224
Data split strategy	3-fold multilabel stratified cross-validation
Main source	ImageCLEFmedical Caption 2026 / ROCOv2-based release

The dataset contains heterogeneous radiology images, including X-ray, CT, MRI, ultrasound, and other imaging appearances. This heterogeneity makes the task more challenging than a single-modality dataset. The sample images shown in Figure 4.1 illustrate this diversity and justify the need for a strong general visual encoder.

The concepts.csv file is the central supervision file. It contains the image identifier and the CUIs associated with that image. A single image may contain one label or several labels. In the development split analyzed in this project, there were no empty-label images, meaning every sample contributed at least one positive concept.

The long-tail distribution is one of the most important dataset characteristics. The most frequent concepts appear in tens of thousands of images, while many concepts have very low support. This affects both learning and evaluation. For this reason, support statistics were computed before model design, and the final model directly uses these statistics to construct frequency groups.

Preprocessing also includes integrity checks. The code verifies that required files exist, that image paths are valid, and that label parsing produces non-empty concept lists. Such checks are important in large challenge datasets because silent file or path errors can lead to misleading results.

4.4 Experiment 1: DenseNet-121 Baseline

The first experimental stage established a strong baseline using DenseNet-121. The model used a sigmoid multi-label head, mixed-precision training, and threshold calibration. This stage demonstrated that DenseNet provides a competitive baseline, achieving a best micro-F1 of 0.5418 at a threshold of 0.475, but it also confirmed that rare concepts remain difficult under a single global thresholding scheme.

Experiment 1 used DenseNet-121 with a multi-label sigmoid classifier. The objective of this stage was not merely to obtain a score, but to establish a baseline strong enough to challenge later models. The model was trained using mixed precision to improve computational efficiency and evaluated through threshold sweeps.

The DenseNet baseline showed that a conventional CNN can achieve competitive micro-F1. However, the long-tail analysis revealed that frequent concepts were much easier than rare concepts. This motivated the search for methods that go beyond a single global classifier and a single global threshold.

Another lesson from the baseline is that the average number of predicted labels per image is highly sensitive to threshold choice. A threshold that predicts too few concepts may produce high precision but low recall, while a threshold that predicts too many concepts may reduce precision. This finding motivated Experiment 2.

4.5 Experiment 2: Threshold Calibration and Cardinality Control

The second stage focused on the decision process rather than the backbone. Using the probabilities generated by the DenseNet baseline, several calibration rules were studied, including micro-opt, balanced, and cardinality-aware calibration. This stage emphasized the importance of matching the predicted label cardinality to the true average number of labels per image. It showed that decision calibration can substantially alter the balance between micro-F1 and semantic richness even when the underlying classifier remains unchanged.

Experiment 2 focused on calibration after model prediction. It used stored prediction probabilities and ground-truth labels to evaluate different decision rules. The micro-opt rule selected the threshold that maximized micro-F1. The balanced rule attempted to balance micro and macro behavior. The cardinality-aware rule attempted to match the average number of predicted labels to the true average of approximately 3.22 labels per image.

This experiment demonstrated that calibration can substantially change the behavior of the model without retraining it. It also showed that optimizing for one metric can harm another. For example, a threshold chosen for macro-oriented behavior may increase recall but produce too many predictions, lowering micro-F1.

The most important methodological lesson is that threshold calibration should be designed together with the final model. This is why the final Swin multi-head model uses per-head threshold calibration rather than relying on one global threshold.

4.6 Experiment 3: SwinGraphHead with Rare/Common Separation

The third stage introduced a transformer-based backbone and explored a rare/common separation strategy. Concepts were divided into rare and common groups and assigned separate thresholds. While this design was conceptually promising, the validation results showed that it was difficult to identify threshold combinations that improved rare-label behavior while preserving the micro-F1 constraint. This stage nevertheless provided valuable insight and motivated a more refined decomposition of the label space.

Experiment 3 introduced a transformer-based model and attempted to separate rare and common concepts. The motivation was correct: rare concepts and common concepts behave differently and should not necessarily share the same decision threshold. However, the two-group split was too coarse.

The calibration search showed that many rare/common threshold pairs could not satisfy the desired micro-F1 constraint while improving rare-label behavior. This indicated that long-tail structure is more gradual than a simple rare/common division. The result motivated a more refined four-group frequency-aware design.

This experiment is important in the thesis because it explains why the final architecture was not chosen arbitrarily. It emerged from an intermediate attempt that revealed the limitations of simpler long-tail handling.

4.7 Final Proposed Model: Swin Frequency-Aware Multi-Head Network

The final proposed model is the main contribution of the thesis. It uses a Swin Transformer backbone (swin_base_patch4_window7_224) to extract hierarchical visual features, then routes these features into four parallel classification heads. Each head is responsible for a subset of concepts grouped according to training frequency: <100 occurrences, 100–500, 500–1000, and >1000. This strategy enables the model to treat rare and frequent concepts differently while still sharing a strong common visual encoder.

Training is performed with Asymmetric Loss, which is more suitable than plain BCE under heavy imbalance. In addition, a WeightedRandomSampler increases the chance of exposing the model to samples containing rare concepts. Exponential moving average (EMA) is used to stabilize the learned weights, and the entire training procedure is evaluated under three-fold multilabel stratified cross-validation. After training, each head undergoes independent threshold calibration. The final prediction is obtained by concatenating the head outputs and applying the calibrated thresholds of the selected operating point.

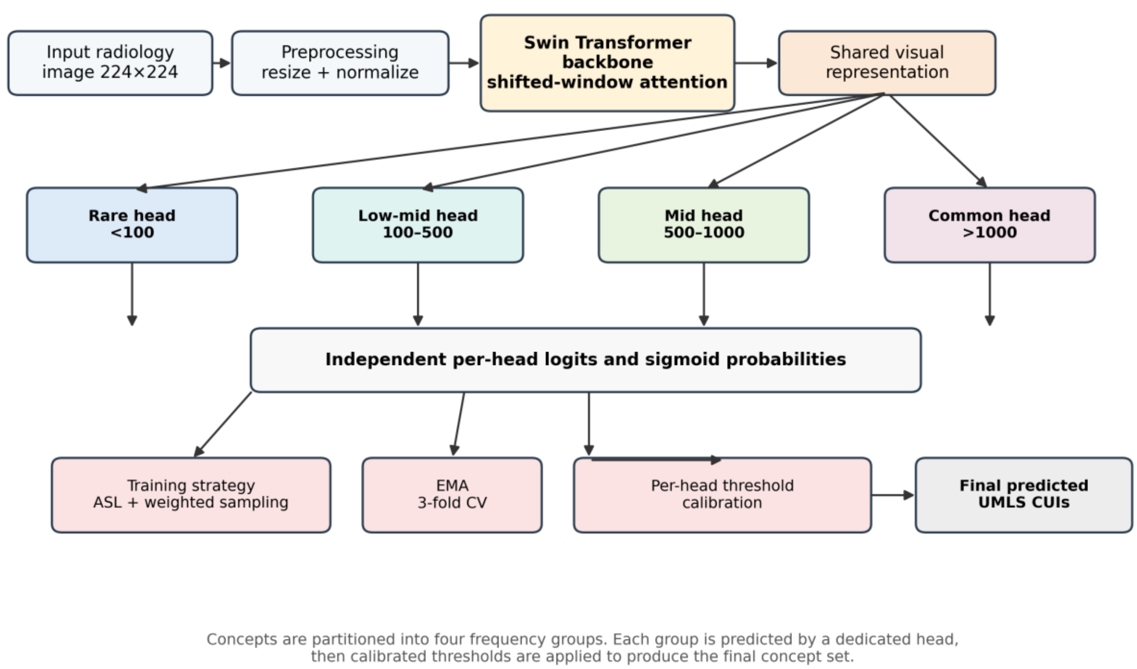

Figure 4.2: Architecture of the final frequency-aware Swin multi-head model.

Table 4.2: Summary of the four main experimental stages.

Stage	Model	Main strategy	Decision rule / note	Micro-F1	Macro_nz	Observation
Experiment 1	DenseNet-121 baseline	Single global threshold	Best threshold = 0.475	0.5418	—	Strong baseline with CNN backbone and hybrid threshold analysis
Experiment 2	DenseNet + calibration	Micro / balanced / cardinality calibration	Cardinality target = 3.22 labels/image	0.5374	0.0079	Improved analysis of label cardinality and threshold behavior
Experiment 3	SwinGraphHead	Rare/common split	Separate thresholds for rare vs. common concepts	0.5405–0.5426	0.0067–0.0092	Explored explicit long-tail handling but without stable gains
Experiment 4	Final Swin frequency-aware multi-head	4 heads by frequency bins	Per-head threshold calibration + ASL + EMA	0.5441 ± 0.0004	0.0073 ± 0.0005	Final proposed model and official submission backbone

The final model is the main experimental contribution. It uses the Swin Transformer backbone to extract visual features and four classification heads to predict different frequency groups. This design combines shared representation learning with specialized output layers.

During training, the model learns all heads jointly. This means the backbone receives gradients from rare, mid-frequency, and common concepts together. However, the final classifier parameters are separated by group, allowing each head to specialize in its subset of the vocabulary.

Asymmetric Loss is used because the target matrix is sparse and contains many easy negatives. The loss down-weights easy negative examples and helps the model focus more effectively on informative labels. WeightedRandomSampler complements this by increasing the presence of samples that carry less frequent concepts.

EMA is used to maintain a smoothed version of the model weights. This often improves validation stability because the averaged weights are less sensitive to short-term fluctuations during training. In a long-tailed multi-label setting, this stability is valuable because rare-label behavior can be noisy.

The final thresholds are not fixed during training. Instead, validation predictions are collected for each fold and thresholds are calibrated after inference. This separation makes it possible to evaluate multiple operating points from the same trained model.

4.7.1 Detailed Architecture Flow

The final model begins by receiving a preprocessed radiology image. The image is resized and normalized according to the Swin backbone requirements. It is then divided internally into patches and processed through hierarchical shifted-window attention blocks. The output of the backbone is a compact visual representation that encodes local and contextual image information.

This representation is shared across all frequency heads. Sharing the backbone is important because rare concepts do not have enough data to train a separate visual encoder. The rare head can therefore benefit from features learned from common and mid-frequency concepts while still having its own classifier parameters.

Each frequency head produces logits only for its assigned concept subset. These logits are passed through sigmoid activations to obtain probabilities. During evaluation, the probabilities from all heads are concatenated in the original vocabulary order so that the final output corresponds to the complete CUI label space.

The four-head design also reduces the conceptual mismatch between common and rare labels. Common concepts may require high thresholds because the model sees many examples and can be more confident. Rare concepts may require different threshold behavior because their scores may be less calibrated due to limited positive examples.

4.7.2 Loss Function and Sampling Strategy

Binary cross-entropy is a natural starting point for multi-label classification, but it can be suboptimal when negative labels dominate the target matrix. In this dataset, each image is positive for only a few concepts and negative for thousands of others. This creates an imbalance within every sample.

Asymmetric Loss modifies the contribution of positive and negative terms. In particular, it reduces the effect of easy negatives, allowing the model to focus more on hard negatives and positive labels. This is especially useful in long-tailed settings where rare positive labels can otherwise be overwhelmed.

WeightedRandomSampler is used to adjust the training sample distribution. Images containing rare labels receive more attention during training, improving the probability that the model observes informative examples for underrepresented concepts. This does not create new data, but it changes exposure frequency in a useful way.

The combination of ASL and sampling addresses imbalance from two different directions: the sampler changes which examples the model sees more often, while the loss changes how errors are weighted. The final model relies on both because either technique alone may be insufficient for a large UMLS vocabulary.

4.7.3 Threshold Calibration and Operating Points

After training, validation inference is performed to collect prediction probabilities. These probabilities are not immediately converted to labels. Instead, a calibration procedure searches for thresholds that produce desirable behavior. This procedure is applied per head in the final model.

The micro-opt operating point selects thresholds that maximize micro-F1. This is useful for challenge evaluation and for comparing models using a single global score. However, it may be conservative and may favor frequent concepts.

The coverage-opt operating point is constrained to remain close to the micro-opt score while improving coverage and macro_nz. This operating point reflects the idea that a concept detector used for caption generation should not only be precise; it should also provide enough semantic content to support descriptive text.

The final model therefore does not have only one fixed interpretation. It can be deployed with different calibrated thresholds depending on whether the application values strict score optimization or richer semantic coverage.

4.8 Chapter Summary

This chapter detailed the methodological progression of the thesis and justified the final design choices. The resulting proposed model is a frequency-aware multi-head Swin network specifically tailored to long-tailed medical concept detection and supported by a calibration strategy that allows multiple useful operating points.

Chapter 5: Research Results

This chapter presents the experimental results obtained from the proposed model and the intermediate experimental stages. It reports validation performance, official submission results, training behavior, and comparisons with related approaches.

5.1 Tools Used and Evaluation Methods

Experiments were developed in Python using PyTorch, timm, pandas, numpy, and matplotlib, and they were executed in a Google Colab Pro environment. Evaluation focused on precision, recall, micro-F1, macro_nz, and concept coverage. These metrics were chosen because they jointly reflect global accuracy, label-level behavior, and semantic breadth. The challenge submission was packaged as a submission.csv file compressed into submission.zip, following the official challenge format.

The experimental environment was designed for reproducibility. Fixed random seeds were used where possible, and the data split was created using multilabel stratification. The use of three folds provided a more reliable estimate of performance than a single random split.

The evaluation metrics were selected to reflect different aspects of model performance. Micro-F1 reflects the main global behavior and is closest to the official score interpretation. Macro_nz provides insight into label-level behavior. Coverage measures whether the model is predicting a diverse set of labels or only a narrow subset of common concepts.

The official submission format required a compressed file containing a submission.csv file with two columns: ID and CUIs. The CUIs were separated by semicolons. The final submission was generated using the final proposed model and successfully accepted by the platform.

5.2 Results of the Proposed Model

The final model achieved a validation micro-F1 of 0.5441 ± 0.0004 at the micro-opt operating point and 0.5405 ± 0.0009 at the coverage-opt operating point. While the coverage-opt setting slightly reduced the global micro-F1, it improved macro_nz by 26.1% and increased concept coverage from approximately 65.7 to 105 concepts. This behavior confirms that the proposed model can be tuned for different practical priorities without requiring retraining.

Table 5.1: Final operating-point results of the proposed model.

Operating point	Micro-F1	Macro_nz	Coverage	Interpretation
Micro-opt	0.5441 ± 0.0004	0.0073 ± 0.0005	65.7	Best global micro-F1 operating point
Coverage-opt	0.5405 ± 0.0009	0.0091 ± 0.0005	105.0	Improves macro_nz by 26.1% with very small micro-F1 drop

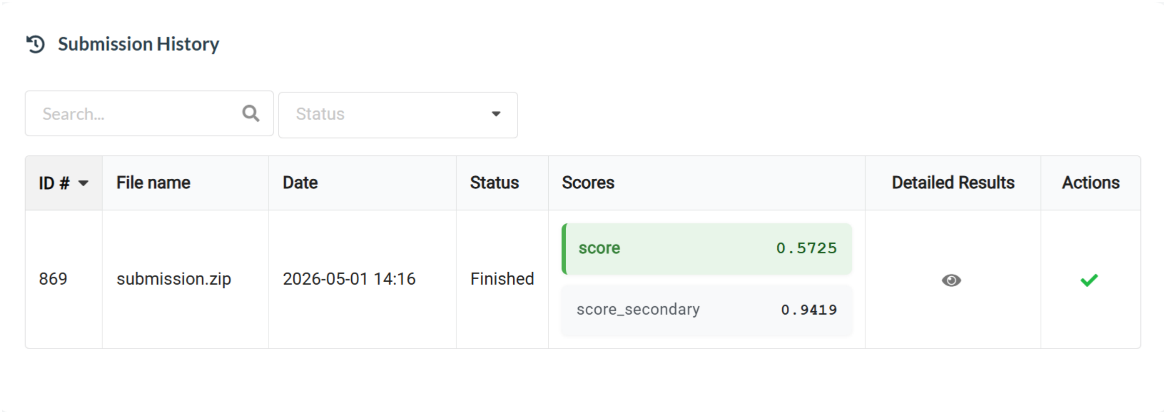

Figure 5.1: Platform-reported official test submission snapshot.

The official challenge submission generated by the final model was successfully accepted by the platform. The submitted run achieved an official test score of 0.5725 and a secondary score of 0.9419. According to the corrected final ranking released by the organizers, our team, basharderar, ranked fourth among the top-performing teams in the Concept Detection Task.

Table 5.2: Corrected final ranking of the ImageCLEFmed Caption 2026 Concept Detection Task.

Rank	Team	Official Score
1	b_wang94	0.5790
2	archimedes	0.5781
3	y_zhang	0.5763
4	quocthai91206	0.5725
4	basharderar	0.5725

The score difference between our submission and the first-ranked team was only 0.0065, indicating that the proposed approach was highly competitive in the official corrected evaluation.

The final model provides two meaningful operating points. The micro-opt operating point is appropriate when the objective is to maximize the main aggregate metric. The coverage-opt operating point is appropriate when the model is intended to support downstream semantic generation, because it produces a broader set of detected concepts.

The improvement in macro_nz under the coverage-opt setting indicates that more concepts receive non-zero predictive behavior. This does not mean that all rare concepts are solved, but it demonstrates that the model can be tuned toward richer semantic output with only a small micro-F1 cost.

The official score of 0.5725 confirms that the final model generalized well from validation to the official test set. The corrected final ranking further shows that the submitted run was competitive among the top teams in the Concept Detection Task.

5.3 Training Dynamics and Operating-Point Analysis

The final training process showed stable convergence across the three folds. The training loss consistently decreased, whereas validation micro-F1 rose quickly in early epochs and then stabilized around the final optimum. Beyond raw training dynamics, the thesis also analyzes the trade-off between selecting the best micro-F1 operating point and selecting a more coverage-oriented operating point. The latter slightly sacrifices micro-F1 in exchange for better macro_nz and broader concept coverage, which is valuable when the downstream application requires richer semantic predictions.

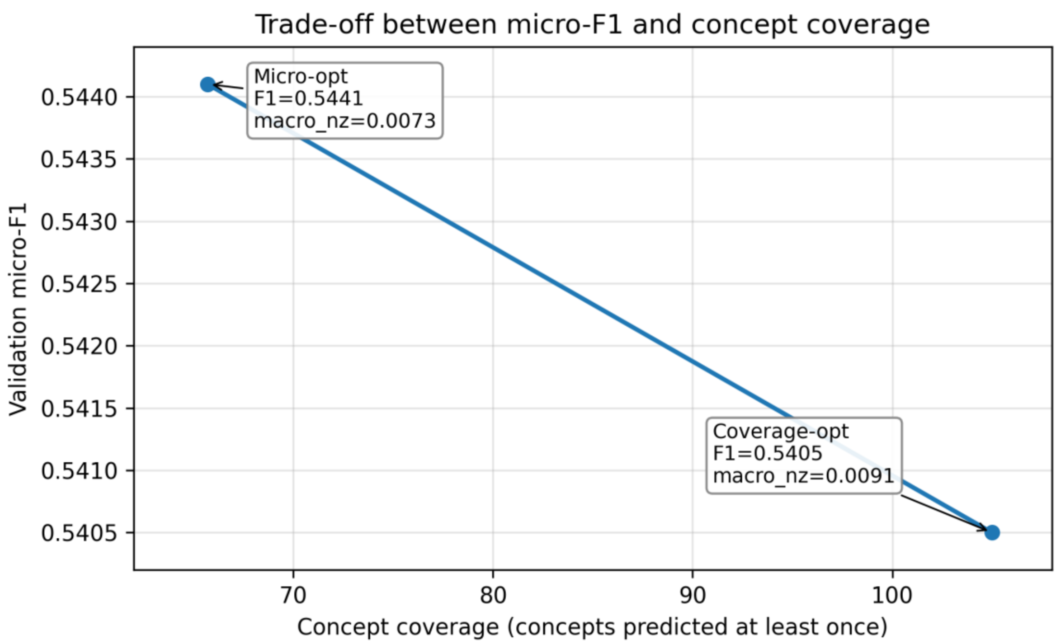

Figure 5.2: Trade-off between micro-F1 and concept coverage.

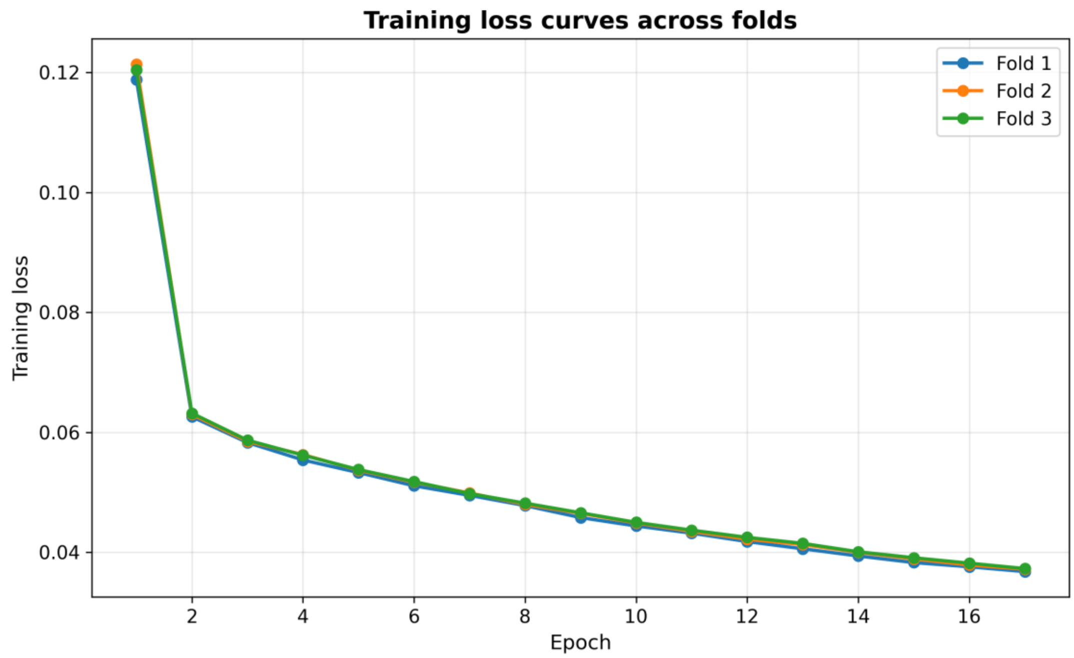

Figure 5.3: Training loss learning curves across the three folds.

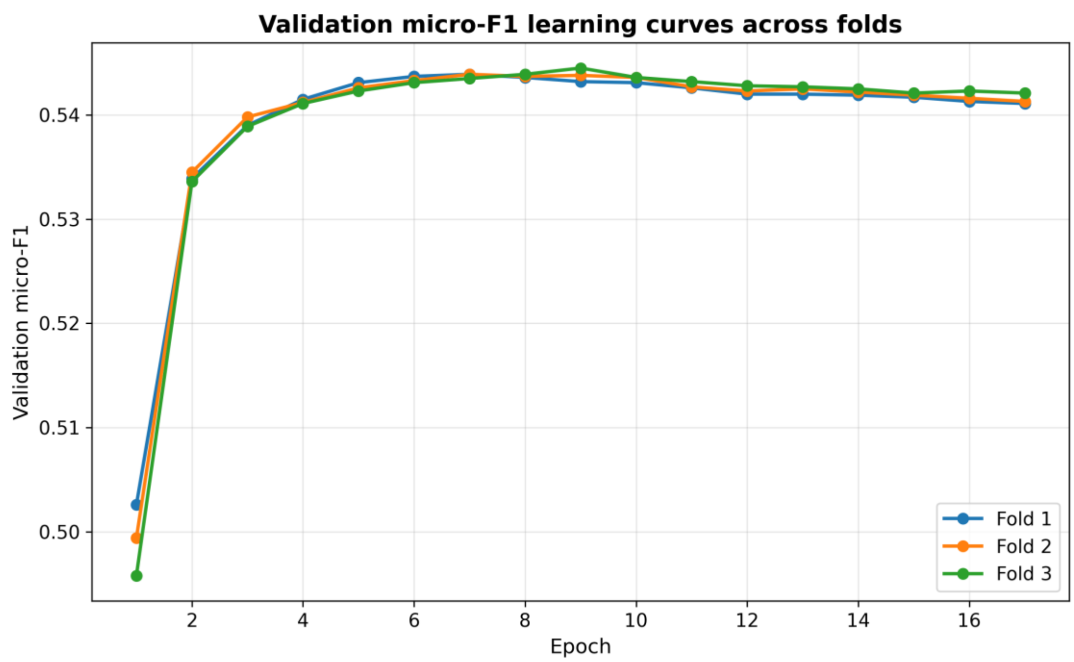

Figure 5.4: Validation micro-F1 learning curves across the three folds.

The learning curves provide evidence of stable optimization. The loss decreases during training, while validation micro-F1 rises and then stabilizes. This behavior indicates that the model learns useful features early and then fine-tunes the decision boundary over later epochs.

The trade-off figure shows why the coverage-opt operating point is important. It increases the number of concepts hit at least once from around 65.7 to approximately 105 while reducing micro-F1 by less than 0.005. This may be useful for downstream caption generation, where semantic coverage can be as important as strict aggregate optimization.

This methodology therefore combines architecture, loss design, sampling, validation, and calibration into a single coherent approach.

The methodology also includes artifact management. Important outputs such as label vocabularies, fold predictions, threshold files, plots, and checkpoints are saved. This is necessary because official submission generation must use the same label order and threshold configuration used during validation. Without careful artifact management, reproducibility would be compromised.

Another methodological concern is computational efficiency. The label space is large and the Swin backbone is relatively heavy. Mixed-precision training and batch-size management are therefore important to make the experiments feasible in a Colab Pro environment. Gradient accumulation and efficient validation batching further support practical training.

The methodology further distinguishes between validation-time analysis and official test inference. Validation uses ground-truth labels and can compute metrics. Test inference does not use labels; it only generates the required predictions. This distinction is important to avoid data leakage and to keep the evaluation scientifically valid.

Finally, the entire pipeline is designed so that the same trained model can produce different calibrated outputs. This makes the model flexible: the micro-opt thresholds can be used for official score optimization, while the coverage-opt thresholds can be studied for richer semantic prediction.

5.4 Comparison of Results

5.4.1 Interpretation of the Final Test Submission

The successful official submission is an important practical validation of the pipeline. It confirms that the generated predictions followed the required format and that the trained model could be applied to the official test images. The official score of 0.5725 provides evidence that the validation behavior translated well to unseen data.

The secondary score of 0.9419 should be interpreted carefully. It is computed under a different filtering condition and is not equivalent to the main score. Nevertheless, the high secondary score indicates that the model performs very well on the subset used by that secondary evaluation.

The difference between validation micro-F1 and the official test score may arise from differences between the validation distribution and the hidden test distribution, the official metric implementation, and the selected operating point. Since the corrected final ranking placed basharderar fourth with a score of 0.5725, the thesis reports the official result as evidence of competitive generalization without overclaiming superiority over all participating systems.

A key strength of the submission is that it was generated by the final proposed model rather than by an ad hoc rule. This means the reported test behavior is directly connected to the methodology described in Chapter 4.

5.4.2 Comparison among the Proposed Models

The staged experimental design made it possible to compare the contributions of different methodological choices. The DenseNet baseline was already strong, which made the task of further improvement non-trivial. Threshold calibration expanded the understanding of operating-point behavior, while the rare/common split experimented with explicit long-tail handling. The final multi-head Swin model provided the best overall balance between global performance and semantic coverage.

Table 5.3: Comparison among the proposed models.

Stage	Model	Main strategy	Decision rule / note	Micro-F1	Macro_nz	Observation
Experiment 1	DenseNet-121 baseline	Single global threshold	Best threshold = 0.475	0.5418	—	Strong baseline with CNN backbone and hybrid threshold analysis
Experiment 2	DenseNet + calibration	Micro / balanced / cardinality calibration	Cardinality target = 3.22 labels/image	0.5374	0.0079	Improved analysis of label cardinality and threshold behavior
Experiment 3	SwinGraphHead	Rare/common split	Separate thresholds for rare vs. common concepts	0.5405–0.5426	0.0067–0.0092	Explored explicit long-tail handling but without stable gains
Experiment 4	Final Swin frequency-aware multi-head	4 heads by frequency bins	Per-head threshold calibration + ASL + EMA	0.5441 ± 0.0004	0.0073 ± 0.0005	Final proposed model and official submission backbone

5.4.3 Comparison with Previous Works

Table 5.4 compares the proposed method with representative prior studies. When compared with challenge-style CNN baselines on concept detection, our final model produces stronger F1 performance. However, some reported results in the literature are obtained on smaller, easier, or reduced concept subsets, and therefore they are not directly comparable to the full long-tail setting targeted in this thesis.

Table 5.4: Comparison between our approach and previous studies.

Method	Setting	Model	Reported result	Comment
Our final model	ImageCLEFmed Caption 2026 validation	Swin frequency-aware multi-head	Micro-F1 = 0.5441	Best internal validation result
Santamato et al. (2025)	ImageCLEF concept prediction	DenseNet121 transfer learning	F1 ≈ 0.26 (3047 concepts)	Our result is substantially higher on our validation protocol
Sahni et al. (2025)	ROCOv2 / ImageCLEF 2025	ResNet50	F1 = 0.4686	Our model outperforms this baseline by a clear margin
Sahni et al. (2025)	ROCOv2 / ImageCLEF 2025	DenseNet121	F1 = 0.4635	Our model outperforms this baseline
Santamato et al. (2025)	Top-20 frequent concepts	DenseNet121 transfer learning	F1 ≈ 0.89	Not surpassed, but the setup is much easier and not directly comparable to full long-tail detection

The proposed method clearly outperforms the challenge-style baselines reported by Sahni et al. (2025) and the full-label setting reported by Santamato et al. (2025). In contrast, it does not surpass the F1 reported by Santamato et al. on the top-20 frequent concepts. This does not indicate a weaker method in general; rather, that particular setting is significantly less difficult because it focuses only on a very small set of common concepts and does not reflect the full long-tail challenge addressed here. Therefore, the fairest conclusion is that our model is particularly strong for the realistic full-vocabulary concept detection setting, while further improvement is still possible for highly simplified frequent-only subsets or under alternative protocols.

The comparison among experiments shows a clear research progression. DenseNet established a strong baseline. Calibration revealed that the decision threshold strongly affects final results. The SwinGraphHead experiment showed that rare/common separation is conceptually useful but insufficient. The final model converted these lessons into a more robust frequency-aware design.

The comparison with previous work must remain cautious because not all studies use the same label set or evaluation protocol. Nevertheless, the proposed model compares favorably with challenge-style CNN baselines and provides a stronger long-tail-aware methodology than simple frequent-label restriction.

The main remaining weakness is rare concept detection. Even with frequency-aware heads and ASL, very rare concepts remain difficult because the model receives few positive examples. This is a fundamental limitation of supervised learning under extreme imbalance and motivates future ontology-aware or language-assisted methods.

5.5 Discussion

The results suggest that improving medical concept detection requires more than selecting a modern backbone. The DenseNet baseline was already strong, so the final gains came from a combination of backbone improvement, frequency-aware structure, loss design, sampling, and calibration. This supports the thesis claim that the problem must be treated as a complete pipeline.

The final model improved the best internal validation micro-F1 while also offering a coverage-oriented operating point. The coverage-opt result is particularly relevant to the broader goal of concept-based caption generation. A caption generator may benefit from a richer concept set even when the strict micro-F1 value is slightly lower.

The long-tail problem remains visible in the results. Frequent concepts are predicted more reliably than rare concepts, which is expected given the distribution of the data. The contribution of this thesis is not to completely solve rare-label detection, but to design a method that explicitly addresses it and improves the trade-off between global accuracy and concept coverage.

The comparison with previous work shows that the proposed model is competitive with challenge-style CNN baselines. However, comparisons across papers should be interpreted cautiously because differences in dataset versions, concept vocabulary size, and evaluation protocols can strongly affect reported scores.

Another important observation is that calibration can change the practical meaning of a model. A model is not fully defined by its architecture and weights; its thresholding rule determines the final set of predicted concepts. This is why the methodology reports calibration details and not only training details.

From an academic perspective, the most valuable outcome is the documented reasoning process. Each experimental stage contributed insight: the baseline showed what a strong CNN can do, calibration showed the importance of inference decisions, the rare/common attempt showed the limitations of coarse label grouping, and the final model showed the benefit of a more structured frequency-aware design.

5.6 Chapter Summary

This chapter reported the experimental environment, validation results, official submission outcome, training dynamics, and comparisons with intermediate models and previous work. The results indicate that frequency-aware multi-head learning improves the final validation behavior and produces a competitive official challenge submission while preserving a clear trade-off between micro-F1 and concept coverage.

Chapter 6: Conclusion

This chapter summarizes the main findings of the project, discusses the remaining challenges, and outlines possible directions for future work. It concludes the thesis by highlighting the value of frequency-aware modeling for long-tailed medical concept detection.

6.1 Challenges and Future Directions

Several challenges remain. First, the long-tailed nature of the label space still limits the detection of rare concepts. Although the proposed frequency-aware heads and calibrated thresholds improve the handling of different frequency ranges, very rare concepts remain difficult because they have limited positive examples. Second, the current approach does not explicitly exploit UMLS ontology relations, even though such relations may provide useful semantic structure. Third, the work focuses on concept detection and does not yet implement the full caption generation stage.

Future work may therefore include ontology-aware graph modeling, concept-aware contrastive learning, stronger hybrid vision-language pretraining, and direct integration of the predicted CUIs into a caption generation model. Additional work may also investigate stronger rare-label augmentation strategies, per-concept calibration, and ensemble methods across folds or backbones.

6.2 Final Conclusion

This thesis investigated the problem of medical concept detection for ImageCLEFmedical Caption 2026 and presented a complete experimental pipeline for long-tailed multi-label radiology concept prediction. The work progressed from a DenseNet-121 baseline to threshold calibration, then to a SwinGraphHead experiment, and finally to a frequency-aware Swin Transformer multi-head model.

The final model achieved the best internal validation performance in the project and produced a successful official test submission. It achieved a validation micro-F1 of 0.5441 ± 0.0004 at the micro-opt operating point and an official score of 0.5725 on the challenge platform, ranking fourth in the corrected final ranking of the Concept Detection Task. These results show that frequency-aware label-space decomposition, Asymmetric Loss, weighted sampling, EMA stabilization, and per-head threshold calibration form an effective strategy for this setting.

In summary, the project demonstrates a complete and academically grounded approach to medical concept detection: it begins with a clear scientific problem, analyzes the dataset and its imbalance, develops several experimental stages, proposes a final architecture, validates the model carefully, and submits the result to an external challenge platform. This makes the thesis a strong foundation for the next stage of concept-based medical caption generation.

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